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ABSTRACT
Regression component analysis (RCA) replaces the factors in a factor analysis model with weighted composites of the model's observed variables. The weight matrix may be calculated from the factor model's parameter estimates. Thus, RCA parameter estimates can be obtained using factor model software, but RCA composites have determinate scores, rather than the indeterminate scores of factors. Analytically, RCA equates to modeling with "regression method" factor scores, except that, while those scores will be inconsistent with the original factor model, they are strictly consistent with the RCA model. When the original factor model is strictly correct in the population and the composites in RCA are standardized, RCA parameter estimates replicate those from regression-weighted forms of partial least squares (PLS) path modeling and generalized structured component analysis (GSCA)—affirming that those methods also equate to modeling with regression method factor scores under the same conditions. Parallel measurement allows RCA to replicate both correlation weight and regression weight versions of PLS and GSCA. These results suggest that RCA and regression-weighted forms of PLS and GSCA are all consistent approaches for modeling data that conforms to a factor model. All analytical methods are described using one consistent symbol palette. Complete R syntax is provided.

KEYWORDS
Structural equation modeling; partial least squares; generalized structured component analysis; regression component analysis; composite-based methods

> This is an instance of a principle that we will encounter repeatedly: the form of a mathematical expression and the way the expression should be evaluated in actual practice may be quite different. Gentle (2017, p. 108)

The common factor model has dominated quantitative social science for decades. Alternative approaches that use weighted composites of observed variables, instead of common factors, have also become popular. With the proliferation of methods, the global body of researchers has fractured into isolated communities, each devoted to their method of choice and using specialized jargon. This hinders scientific communication in many ways. Research produced by members of a given community, using idiosyncratic methods and terminology, may be unintelligible outside the community. It may be easily dismissed as flawed or irrelevant, thus sacrificing information. Understanding a range of methods in the same terms may help researchers fit a given tool to a given problem and avoid falling prey to groupthink.

Across several papers published during the 1970s, Peter Schönemann and coauthors (e.g., Schönemann, 1971; Schönemann & Steiger, 1976; Schönemann & Wang, 1972) explored the phenomenon of factor indeterminacy. Among these contributions, Schönemann and Steiger (1976) described regression component analysis (RCA). This analytical method starts with a common factor model but replaces the indeterminate common factors with determinate weighted composites of the observed variables. What was once a common factor model is transformed into a model of weighted composites.

During the years since, partial least squares (PLS) path modeling (Rigdon, 2013; Wold, 1980, 1981) and generalized structured component analysis or GSCA (e.g., Hwang & Takane, 2004, 2015), in particular, have gained prominence as methods based on weighted composites rather than on common factors. Today, Schönemann and Steiger's (1976) RCA, firmly rooted in common factor analysis, offers a valuable window into these later composite-based methods.

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This paper aims to illustrate connections between these methods, using RCA to bridge a gap in understanding and thus establish a basis for more fruitful dialogue.

This presentation first briefly reviews the common factor model and the factor indeterminacy phenomenon which inspired Schönemann and Steiger (1976). Then this presentation treats the RCA model and the estimation of its parameters, with a particular focus on the connection between loadings and weights. The following sections review model structure and parameter estimation for PLS path modeling and GSCA. After illustrating the relationships between those analytical frameworks, this presentation considers empirical examples. The presentation concludes with broader implications for composite-based approaches to structural equation modeling. This presentation will use one consistent symbol palette to clarify understanding of all methods discussed.

The common factor model and factor indeterminacy

A common factor model for P mean-centered observed variables $\mathbf{y}_p$, $p = 1$ to P, may be represented as:

$$\mathbf{y} = \boldsymbol{\Lambda}\boldsymbol{\eta} + \boldsymbol{\varepsilon} \tag{1}$$

Here, $\mathbf{y}$ is a $P \times 1$ vector of observed variables, $\boldsymbol{\Lambda}$ ("Lambda") is a $P \times K$ matrix of loadings, $\boldsymbol{\eta}$ ("eta") is a $K \times 1$ vector of common factors, and $\boldsymbol{\varepsilon}$ ("epsilon") is a $P \times 1$ vector of specific factors or error terms. (As with all methods described here, the "structural part" of each model, which links one common factor or composite to another, will not be presented.) Consistent with standard regression assumptions, the specific factors $\boldsymbol{\varepsilon}$ are typically specified as independent of the common factor predictors, $\boldsymbol{\eta}$. If the common factors and specific factors all have means of 0, then this model implies the covariance structure:

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Phi}\boldsymbol{\Lambda}' + \boldsymbol{\Psi} \tag{2}$$

Here, $\boldsymbol{\Sigma}$ ("Sigma") is the $P \times P$ covariance matrix of the observed variables, $\boldsymbol{\Phi}$ ("Phi") is the $K \times K$ covariance matrix of the common factors, $\boldsymbol{\Psi}$ ("Psi") is the $P \times P$ covariance matrix of the specific factors, and $\boldsymbol{\Lambda}'$ indicates the matrix transpose of $\boldsymbol{\Lambda}$. In classical common factor analysis, $\boldsymbol{\Psi}$ is taken to be diagonal. However, the method allows researchers flexibility to include non-zero off-diagonal values, representing covariance between observed variables that is not mediated by the common factors.

What makes the $\boldsymbol{\eta}$ variables common factors and not determinate functions of the observed variables is most easily seen by creating one combined covariance matrix for the $\boldsymbol{\eta}$ and $\boldsymbol{\varepsilon}$ variables. Given the assumed independence of common and specific factors, the combined covariance matrix $\mathbf{Q}$ is a partitioned block diagonal matrix:

$$\mathbf{Q} = \begin{bmatrix} \boldsymbol{\Phi} & \mathrm{O} \\ \mathrm{O} & \boldsymbol{\Psi} \end{bmatrix} \tag{3}$$

Here $\mathbf{O}$ indicates a submatrix of 0s of suitable dimension. The issue that sets common factor models apart is the rank of $\mathbf{Q}$. If the common factors and the specific factors are mutually orthogonal and the covariance matrices $\boldsymbol{\Phi}_{KxK}$ and $\boldsymbol{\Psi}_{PxP}$ both have full rank, then the rank of $\mathbf{Q}$ in Equation (3) will be $P + K$.

The problem is that $\boldsymbol{\Sigma}$, the covariance matrix of the P observed variables on the left-hand side of Equation (2)—which represents all of the information available to use in estimating the model—can have rank no more than P. The common factor model turns P things into $P + K$ things. Put another way, if one wants to construct values for the $\boldsymbol{\eta}$ and $\boldsymbol{\varepsilon}$ variables out of the information in the $\mathbf{y}$ variables, there is not enough information to go around. Building the $P + K$ $\boldsymbol{\eta}$ and $\boldsymbol{\varepsilon}$ variables only using the P $\mathbf{y}$ variables would involve using the same information more than once, and this re-use of information would create dependencies across the $\boldsymbol{\eta}$ and $\boldsymbol{\varepsilon}$ variables, in conflict with the independence and full rank assumptions in the common factor model.

Avoiding that redundancy means creating new information, independent of what exists in the original data, using a formula (e.g., Rigdon et al., 2019, p. 430, Equation (6)):

$$\boldsymbol{\eta}_{Kx1} = \boldsymbol{\Phi}\boldsymbol{\Lambda}'\boldsymbol{\Sigma}^{-1}\mathbf{y} + \mathbf{B}_{KxK}\mathbf{z}_{Kx1} \tag{4}$$

$\mathbf{B}_{KxK}$ is a matrix of weights, $\boldsymbol{\Sigma}^{-1}$ indicates the matrix inverse of $\boldsymbol{\Sigma}$, and $\mathbf{z}$ is a set of K arbitrary variables z_k, $k = 1$ to K, one per common factor, each with variance equal to that of its associated common factor. Each z_k is required to be orthogonal to all variables in the model except the common factor of which it forms a part.

There is no unique best set of values for the $\mathbf{z}$ variables, and hence no one best set of scores on the common factors. The infinitely various possible sets of scores need not maintain rank order across cases, and different sets of scores for the same factor need not even be positively correlated (Guttman, 1955; Rigdon et al., 2019). So, for example, if one were creating scores on job performance to rank employees, the factor model may not produce one best set of rankings (unless indeterminacy is small), and employees who are disadvantaged by a particular ranking or relative score might question why that particular set of partly arbitrary scores was used. For some researchers (Guttman, 1955; Rigdon et al., 2019), this indeterminacy left open questions about what, exactly, a given common factor represented. Other researchers note that factor indeterminacy has no direct effect on parameter estimation or goodness-of-fit tests, and so are inclined to ignore the phenomenon[1].

There are different ways to eliminate or at least reduce factor indeterminacy in a model, though eliminating indeterminacy necessarily turns the common and specific factors into composites. For example, for each common factor in the model, one could fix to zero the variance of one specific factor (i.e., error) (Krijnen et al., 1998). This would reduce the rank of $\mathbf{Q}$ from $P+K$ down to P, the same as the rank of the observed variable covariance matrix. Alternatively, "confirmatory composite analysis" (Henseler, 2021) and the "Henseler-Ogasawara specification" (Schuberth, 2023; Yu et al., 2023) replace the P specific factors with $P-1$ common factors with restricted patterns of loadings. These approaches again reduce the total number of unobserved variables, and the rank of $\mathbf{Q}$, to P.

Alternatively, one might reduce indeterminacy by adding more and better indicators to the model. The degree of indeterminacy shrinks toward 0 as the number of indicators per common factor and the strength of loadings increase. At the limit, the weights in the $\mathbf{B}$ matrix in the second term in Equation (4) approach 0. However, data collection may become much more onerous and expensive.

One might construct scores that omit the arbitrary component. The "regression method" for computing such scores does exactly this (e.g., Mulaik, 2010, p. 375, Equation 13.15), using a formula that omits the $\mathbf{Bz}$ term from Equation (4). Such scores, however, will not be entirely consistent with the original common factor model. The $\mathbf{z}$ variables in these terms are arbitrary and orthogonal, so excluding the $\mathbf{Bz}$ terms from the factor score calculations will not affect covariances among the approximate factors scores. But the exclusion of the $\mathbf{Bz}$ terms will produce approximate factor scores with smaller variances and larger correlations than what the estimated factor model indicates for the common factors themselves. This discrepancy could leave researchers doubting the validity of any subsequent interpretations that rely on those scores. Alternatively, one might directly employ an analytical method using weighted composites with determinate values, instead of indeterminate factors. The latter approach might stand as one consistent and auditable process.

Regression component analysis

Schönemann and Steiger (1976) described a form of the latter approach for avoiding factor indeterminacy. Their approach replaced both common factors and specific factors with weighted composites of the model's observed variables. Their "regression component decomposition" could be represented as:

$$\mathbf{y} = \mathbf{\Lambda W'y} + (\mathbf{I} - \mathbf{\Lambda W'})\mathbf{y} \tag{5}$$

Here $\mathbf{W}_{PxK}$ is a matrix of weights, and $\mathbf{I}$ is an identity matrix of suitable dimension, here P x P. (Notice that the weight matrix is transposed in Equation (5). The weight matrix $\mathbf{W}$ itself has the same dimension as $\mathbf{\Lambda}$, the loading matrix.) The weighted composites $\mathbf{W'y}$ replace the common factors $\boldsymbol{\eta}$. The loading matrix $\mathbf{\Lambda}$ represents the regression of the observed variables on their associated composites. If a researcher already had a set of weights, they could calculate loadings as (Schönemann & Steiger, 1976, p. 178, Equations 1.5 and 1.5a):

$$\begin{gathered}\mathbf{\Lambda} = \mathrm{cov}(\mathbf{y}, \mathbf{W'y})/\mathrm{var}(\mathbf{W'y}) \\ \mathbf{\Lambda} = \mathbf{\Sigma W \Phi}^{-1}\end{gathered} \tag{6}$$

where $cov(a,b)$ indicates the covariance of a and b, and $var(a)$ indicates the variance of a. However, the factor model user could obtain their loadings by estimating a factor model, using their factor analysis software of choice (Schönemann & Steiger, 1976, p. 179).

Alternatively, given the parameters of a factor model, the weight matrix $\mathbf{W}$ can be derived as (Schönemann & Steiger, 1976, p. 179, Equation 2.2):

$$\begin{gathered}\mathbf{\Lambda} = \mathbf{\Sigma W \Phi}^{-1} \\ \mathbf{\Sigma}^{-1}\mathbf{\Lambda \Phi} = \mathbf{W} \\ \mathbf{\Sigma}^{-1}\mathbf{\Lambda}(\mathbf{\Lambda}'\mathbf{\Sigma}^{-1}\mathbf{\Lambda})^{-1} = \mathbf{W}\end{gathered} \tag{7}$$

since it can be shown that:

$$\mathbf{\Phi} = \mathbf{W'\Sigma W} = (\mathbf{\Lambda \Sigma}^{-1}\mathbf{\Lambda})^{-1} \tag{8}$$

(Schönemann & Steiger, 1976, p. 179). With the results of a factor analysis in hand, RCA makes it straightforward to replace the indeterminate common factors with weighted composites, while changing the model itself to be a model[2] of composites rather than a model of factors approximated by composites.

[1]One highly accomplished researcher, who was leading a workshop on exploratory factor analysis, characterized factor indeterminacy as "a nothingburger.".

[2]An editor points out that Equation (5), by itself is only a decomposition and not a falsifiable model. Schönemann and Steiger (1976, pp. 185–186),

Regression weights vs correlation weights

Note from the second line of Equation (7) that:

$$\mathbf{W}'\mathbf{y} = (\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda}\boldsymbol{\Phi})'\mathbf{y} = \boldsymbol{\Phi}\boldsymbol{\Lambda}'\boldsymbol{\Sigma}^{-1}\mathbf{y} \tag{9}$$

which shows that RCA produces scores for its composites, $\mathbf{W}'\mathbf{y}$, using the same formula as that for "regression method" factor scores—the first product on the right-hand side of Equation (4). Such scores are not perfectly consistent with the common factors in a factor model, but they are perfectly consistent with the composites in regression component analysis.

The RCA weights are, indeed, regression weights, as opposed to "correlation weights" (Waller & Jones, 2010). Correlation weights are weights that ignore collinearity among predictors. Here, the original observed variables $\mathbf{y}$ are the predictors and $\boldsymbol{\Sigma}$ in Equation (9) captures their (model-implied) covariance.[3] Regression weights are optimal in terms of maximizing variance explained within the dataset used to estimate the weights. However, they are not necessarily optimal in terms of variance explained in the larger population from which the dataset was randomly sampled (Dana & Dawes, 2004). Correlation weights will be superior in out-of-sample prediction under some fairly plausible conditions, and ignoring collinearity avoids the interpretational confusion due to phenomena like "suppression" (Cohen et al., 2003, pp. 77–78) or inflation of estimated standard errors (Cohen et al., 2003, pp. 419–430).

Transformation and standardization

With the product $\boldsymbol{\Lambda}\mathbf{W}'$ in both terms on the right-hand side of Equation (5), a researcher could rescale those parameters without effecting any substantive change in the solution, using any nonsingular transformation matrix $\mathbf{T}_{KxK}$, for example *via*:

$$\mathbf{y} = \boldsymbol{\Lambda}\mathbf{T}\mathbf{T}^{-1}\mathbf{W}'\mathbf{y} + (\mathbf{I} - \boldsymbol{\Lambda}\mathbf{W}')\mathbf{y} \tag{10}$$

or

$$\mathbf{y} = \boldsymbol{\Lambda}\mathbf{W}'\mathbf{y} + (\mathbf{I} - \boldsymbol{\Lambda}\mathbf{T}\mathbf{T}^{-1}\mathbf{W}')\mathbf{y} \tag{11}$$

with their Theorems 4 and 5, showed an equivalence between a correct factor model and a constraint on the covariance matrix of the error terms in RCA. The constraint specifies that the covariance matrix can be rescaled to a diagonal idempotent matrix with P – K 1's on the diagonal and 0's everywhere else. With this constraint in place, RCA is a falsifiable model, and will be falsified if and only if the original factor model is falsified.

[3]For consistency in calculations, Σ should be the model-implied covariance matrix from the factor analysis that produced the other parameter estimates. In the absence of sampling variance and factor model misspecification, the empirical covariance matrix can be used equivalently.

or with any nonsingular matrix $\mathbf{T}_{PxP}$:

$$\mathbf{y} = \mathbf{T}\boldsymbol{\Lambda}\mathbf{W}'\mathbf{T}^{-1}\mathbf{y} + (\mathbf{I} - \boldsymbol{\Lambda}\mathbf{W}')\mathbf{y} \tag{12}$$

Indeed, different $\mathbf{T}$ matrices might be used for the first and second terms on the right-hand side. Such an approach might be used to standardize the $\mathbf{W}'\mathbf{y}$ composites *via* $\mathbf{T}^{-1}\mathbf{W}'\mathbf{y}$, where $\mathbf{T}$ might be a diagonal matrix containing the standard deviations of the composites $\mathbf{W}'\mathbf{y}$ (Schönemann & Steiger, 1976, p. 179):

$$\mathbf{T} = \mathrm{diag}(\mathbf{W}'\boldsymbol{\Sigma}\mathbf{W})^{\frac{1}{2}} = \mathrm{diag}(\boldsymbol{\Lambda}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda})^{-\frac{1}{2}} \tag{13}$$

if such standardization were desired, and also replacing $\boldsymbol{\Lambda}$ with $\boldsymbol{\Lambda}\mathbf{T}$. Standardization is possible but not required in either factor-based models or in regression component analysis, neither for the factors/composites nor for the observed variables.

Assumptions and fit

Schönemann and Steiger (1976, p. 179) noted that regression component analysis, like common factor analysis, yields the result that $\boldsymbol{\Psi} = \boldsymbol{\Sigma} - \boldsymbol{\Lambda}\boldsymbol{\Phi}\boldsymbol{\Lambda}'$, as long as $\boldsymbol{\Lambda}'\mathbf{W} = \mathbf{I}$, in other words, as long as $\boldsymbol{\Lambda}$ is the "right inverse" of $\mathbf{W}'$, or equivalently, as long as $\mathbf{W}'$ is the "left inverse" of $\boldsymbol{\Lambda}$ (Gentle, 2017, p. 108). This is true in RCA—using Equations (6)–(8) (Schönemann & Steiger, 1976, pp. 178–179):

$$\mathbf{W}'\boldsymbol{\Lambda} = \mathbf{W}'\boldsymbol{\Sigma}\mathbf{W}(\mathbf{W}'\boldsymbol{\Sigma}\mathbf{W})^{-1} = \mathbf{I} \tag{14}$$

again, depending on the assumption that the common factor model is correct. Thus, regression component analysis yields a structure comparable to the factor-based model, with straightforward estimation and without indeterminate common factors. If the original factor model is correct, then the RCA model is also correct for the same population. RCA produces a composite-based model consistent with that same population and composite scores matching the estimated model. RCA's reliance on a correct factor model suggests that labeling it a "composite method" is not entirely correct, while labeling RCA a "quasi-factor method" captures the method more comprehensively.

Partial least squares path modeling

Partial least squares (PLS) path modeling began to emerge at about the same time as Schönemann and Steiger's (1976) regression component analysis (Noonan, 2017; Wold, 1980, 1981), but with a completely different purpose. The design of the PLS path modeling approach aimed to minimize requirements in terms of prior knowledge, data and (in that era) computing resources, in reaction to the relatively high demands associated with common factor analysis. PLS path modeling's design sacrificed elegance and statistical optimality in favor of practical utility (Wold, 1981).

The typical PLS path modeling algorithm first attempts to develop weights $\mathbf{W}$ such that weighted composites $\mathbf{W}'\mathbf{y}$ maximize correlations across those composites which are directly connected by paths in a structural model. Observed variables are typically standardized, so that one observed variable with outsized variance will not dominate a composite merely because of its scaling. Standardization of the composites—choosing weights that produce composites with unit variance—resolves an identification problem. The estimation process involves alternately replacing composites formed from observed variables with composites formed from other composites in the model. When this process achieves convergence, the result is a correlation matrix that includes both the original observed variables and weighted composites of those observed variables, which stand in the place of the common factors in a common factor model. Path models in PLS path modeling are typically recursive, so that least squares estimation is sufficient to estimate both loadings and structural parameters from the comprehensive correlation matrix.

Relations between observed variables and composites predominantly take one of two forms, typically "Mode A" and "Mode B." Here, the two approaches will be labeled "PLS_A" and "PLS_B." Conventionally, PLS_A incorporates loadings into the model, while PLS_B does not. Thus, PLS_A implies a model of the form:

$$\mathbf{y} = \boldsymbol{\Lambda}\mathbf{W}'\mathbf{y} + \mathbf{e} \tag{15}$$

while PLS_B excludes both loadings and error terms. In any event, loadings as zero-order correlations are easily found in the comprehensive correlation matrix once the algorithm converges on a set of weights. In that sense, weights and other model parameter estimates define the loadings. Relations among the error terms in Equation (15) are unconstrained, for those observed variables associated with the same composite.

If each observed variable is associated with only one composite, then under Mode A, the weights used to produce a weighted composite in PLS path modeling are equivalent to correlation weights (Rigdon, 2013, pp. 90–91), which ignore collinearity. The standardization of both the observed variable and composite, and the unconstrained error terms on the observed variable side, mean that a regression of one observed variable on a composite is equivalent to a regression of one composite on one observed variable.

However, if a dataset is drawn from a population where a common factor model is correct, then the observed variables—the predictors in a method of composites—should be correlated, preferably highly correlated. Correlation weights will thus fail to reproduce the factor model-defined population in the general case (Dijkstra & Henseler, 2015).

Generalized structured component analysis

Citing PLS path modeling's reliance on a set of algorithms rather than on one overall optimization criterion, Hwang and Takane (2004, 2015) presented an alternative to PLS path modeling employing a single overall criterion. Still, like PLS path modeling, this overall criterion aims to explain variance, rather than the typical factor analysis aim of matching an empirical covariance matrix. The GSCA approach includes a "weighted relations model," which shows how the K composites, $\boldsymbol{\gamma}$, are weighted sums of the P observed variables (e.g., Cho & Choi, 2020):

$$\boldsymbol{\gamma} = \mathbf{W}'\mathbf{y} \tag{16}$$

where $\boldsymbol{\gamma}$ is a $K \times 1$ vector of composites. In this presentation, this form of the GSCA model will be labeled "GSCA_W" using W for "weights."

There is also an optional "measurement model" where the observed variables are predicted by the composites, using loadings:

$$\mathbf{y} = \boldsymbol{\Lambda}\boldsymbol{\gamma} + \mathbf{e} \tag{17}$$

Like PLS path modeling, GSCA does not depend on a factor model being correct in the population. Unlike PLSc, which uses a transformation of Mode A weight estimates to obtain estimates of factor model loadings, GSCA uses a different model form, labeled "GSCAm," which incorporates "uniqueness terms" (Hwang et al., 2017). Like RCA and PLS path modeling, GSCA yields determinate scores on the composites for each case in the dataset.

Equations (16) and (17) can be combined to obtain:

$$\mathbf{y} = \boldsymbol{\Lambda}\mathbf{W}'\mathbf{y} + \mathbf{e} \tag{18}$$

In this presentation, this form of the GSCA model will be labeled "GSCA_WL," where "$_\text{WL}$" represents "weights and loadings."

Comparing Equation (18) to Equation (5), GSCA_WL and RCA are equivalent in form as long as:

$$\mathbf{e} = (\mathbf{I} - \boldsymbol{\Lambda}\mathbf{W}')\mathbf{y} \tag{19}$$

which is true from Equation (18):

$$\begin{aligned}\mathbf{y}-\boldsymbol{\Lambda}\mathbf{W}'\mathbf{y}&=\mathbf{e}\\(\mathbf{I}-\boldsymbol{\Lambda}\mathbf{W}')\mathbf{y}&=\mathbf{e}\end{aligned}\tag{20}$$

However, this form also appears to be equivalent to PLS_A, which differs from RCA because PLS_A weights are actually correlation weights while RCA weights are regression weights. The equivalence in form between $GSCA_{WL}$ and PLS_A actually suggests that the weights used to form composites in $GSCA_{WL}$ are also correlation weights, while weights in $GSCA_W$ are regression weights, the same as in RCA. In any event, barring advanced constraints, loadings in a composite-based method play no role in shaping the composites. The composites are defined by the observed variables and the weights. If the dataset is the same and the weights are the same, then, necessarily, the composites will be the same, and then the loadings or correlations between observed variables and composites will also be the same.

Hwang and Takane (2004, 2015) proposed estimating the parameters of their model using a multistep alternating least squares algorithm. In each iteration of the estimation process, weights are updated, then loadings (and structural parameters linking the composites) are updated. The estimation routine terminates when, effectively, the change in parameter estimates from round to round is less than some specified quantity. This approach allows the possibility of imposing constraints on loadings, such as equality constraints or constraints that instantiate a latent growth model in factor analysis. Like PLS path modeling, GSCA builds its composites out of standardized observed variables, and includes constraints on $\mathbf{W}$ that standardize the composites, $\mathbf{W}'\mathbf{y}$.

Empirical examples

Empirical examples illustrate similarities and differences in the performance of this set of analytical techniques, and connections between those techniques (see Table 1). At the level of the finest distinctions, this analysis encompasses seven different techniques. Of those, factor analysis with standardized common factors and regression component analysis retaining that factor-based standardization ("RCA_F") have identical parameter estimates, except that the latter method additionally includes a matrix of weights. Regression component analysis with standardized composites ("RCA_C"), rather than common factors, has the same form as RCA_F but with different parameter values. The other four techniques are partial least squares path modeling with loadings ("PLS_A") and without loadings ("PLS_B") and generalized structured component analysis with loadings ("$GSCA_{WL}$") and without loadings ("$GSCA_W$").

Table 1. Analytical techniques and empirical examples.

Analytical Techniques
FA: Confirmatory factor analysis with factor standardization scaling
RCA_F: Regression component analysis retaining factor standardization scaling
RCA_C: Regression component analysis with standardized composite scaling
PLS_B: Partial least squares mode B (weights only)
$GSCA_W$: Generalized structured component analysis with weights only
PLS_A: Partial least squares mode A (weights and loadings)
$GSCA_{WL}$: Generalized structured component analysis with weights and loadings
Empirical Examples
1. Zero discrepancy factor model population with parallel indicators
2. Zero discrepancy factor model population with congeneric indicators
3a. Zero discrepancy composite model population
3b. Example 3a converted to a zero discrepancy factor model population with congeneric indicators
4a. Real data, finite sample size, imperfect fit to a factor model
4b. Example 4a converted to a zero discrepancy factor model population with finite sample size and congeneric indicators

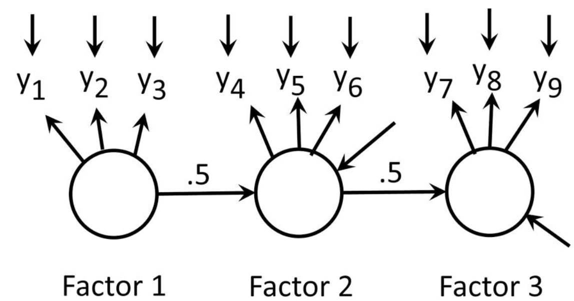

Figure 1. Path model for empirical examples 1, 2 and 3.

Empirical example 1: simulated data with parallel indicators

Method

The first empirical example illustrates the peculiar equivalence associated with a factor population with *parallel* indicators. The model (see Figure 1) has three correlated common factors, each associated with three observed variables. The model's population parameter values are fully standardized. The structural model involves direct paths with values of 0.5 between factors 1 and 2 and between factors 2 and 3, with no direct path between factors 1 and 3, implying a factor correlation matrix of the form:

$$\boldsymbol{\Phi}=\begin{bmatrix}1&&\\.50&1&\\.25&.50&1\end{bmatrix}\tag{21}$$

Loadings are identical for all indicators for each common factor (all 0.9 for the first common factor, all 0.8 for the second, and all 0.6 for the third). These values were chosen to illustrate the impact of equivalence across a range of likely loading values. Variances of specific factors are set to imply unit variance for each observed variable, making the indicators parallel for each common factor.

Data were created in R (version 4.3.0) by first specifying values for the $\mathbf{\Phi}$ and $\mathbf{\Psi}$ matrices from Equation 2, given the specification described here and in Figure 1[4]. Then the mvrnorm() function in the R MASS package (version 7.3-54) was used to generate 50,000 multivariate normal observations on the common and specific factors. The high sample size was chosen to minimize rounding error (Bedeian et al., 2009). To further clarify the empirical relationships, the data were adjusted to remove all random sampling variance by applying a Bollen-Stine type adjustment (Bollen & Stine, 1992). If $\hat{\mathbf{\Sigma}}$ and $\bar{\mathbf{x}}$ are the sample covariance matrix and sample mean vector of a dataset $\mathbf{D}_{old}$, and $\mathbf{\Sigma}$ is the population covariance matrix from which $\mathbf{D}_{old}$ was sampled, then:

$$\mathbf{D}_{new} = (\mathbf{D}_{old} - \bar{\mathbf{x}})\hat{\mathbf{\Sigma}}^{-\frac{1}{2}}\mathbf{\Sigma}^{\frac{1}{2}} \tag{22}$$

transforms the original dataset into a new dataset, $\mathbf{D}_{new}$, whose variables all have means of exactly 0 and whose covariance matrix *exactly* matches the population covariance matrix, $\mathbf{\Sigma}$, rather than $\hat{\mathbf{\Sigma}}$. Finally, using the specified loadings, values for the common and specific factors were combined to create the observed variables, consistent with Equation (1). Supplementary materials contain annotated R syntax for each of the paper's empirical examples.

The lavaan package (version 0.6-9) was used to estimate the common factor model in Figure 1 using these data. A weight matrix $\mathbf{W}$ was calculated using Equation (7). Scores for the composites—the weighted sums that stand in place of common factors in RCA—were computed using the data and the weights, as $\mathbf{W}'\mathbf{D}_{new}$ (see Table 2).

Using the same simulated data, PLS path models were estimated using the cSEM R package (version 0.4.0), and GSCA models were estimated using both the cSEM package and Hwang et al., (2021) GSCAPro (version 1.1) stand-alone software, in order to check for discrepancies. This study obtained results for both PLS_A and PLS_B, and both GSCA_WL and GSCA_W.

[4]While this analysis could have been conducted at the level of population moments, as an editor points out, factor indeterminacy is a phenomenon at the level of individual observations (Schönemann & Steiger, 1976). Therefore, this analysis proceeds using individual observations.

Table 2. Parameter estimates from empirical example 1: Parallel indicators.

	Regression component analysis			GSCA and PLS path modeling	
	FA	RCA_F	RCA_C	$\text{PLS}_\text{B}/\text{GSCA}_\text{W}$	$\text{PLS}_\text{A}/\text{GSCA}_\text{WL}$
Loadings					
y1, y2, y3	0.900		0.935	0.935	0.935
y4, y5, y6	0.800		0.872	0.872	0.872
y7, y8, y9	0.600		0.757	0.757	0.757
Weights					
y1, y2, y3		0.370	0.357	0.357	0.357
y4, y5, y6		0.417	0.382	0.382	0.382
y7, y8, y9		0.556	0.440	0.440	0.440

FA: factor analysis; RCA_F: regression component analysis with factor standardization; RCA_C: regression component analysis with composite standardization; PLS_B: partial least squares path modeling mode B; GSCA_W: generalized structured component analysis with weights only; PLS_A: partial least squares path modeling mode A; GSCA_WL: generalized structured component analysis with weights and loadings.

Results

Given that the factor model is strictly correct in the population and that all sampling variance is removed, the returned factor model parameter estimates exactly match the programmed values. In the regression component analysis results, the *covariances* among the weighted composites match the *correlations* assigned to the common factors (see Table 2). However, variances differ, so correlations among the composites do not match the common factor correlations. Remember that RCA aims not to mimic a factor model but rather to describe a population using weighted composites.

The initial (unstandardized) RCA results differ from PLS and GSCA results. The differences are entirely due to the standardization built into GSCA and PLS path modeling. After applying the standardization transformation for the RCA composites (Equation (12)), all results from RCA analysis are identical to results from GSCA and PLS path modeling. This distinguishes the impact of replacing common factors with weighted composites from the impact of standardizing the composites, on the parameter estimates. Recall that, without the standardization of composites, RCA employs the original factor model parameter estimates. Presumably, the other methods would behave similarly if it were possible to omit standardization from their estimation processes.

Also, note that PLS and GSCA results are identical regardless of specific form. This is partly due to the required standardization of composites, but also particularly due to the observed variables being parallel indicators, in this very artificial case, and to the correct factor model that underlies the data. Given the strictly correct factor model and standardization, equal loadings for a set of indicators imply equal correlations for those indicators across all other variables in the model (Jöreskog, 1979). For methods like PLS path modeling and GSCA that seek to optimize correlations between variables, this situation offers no opportunity to improve results by favoring one observed variable over another in forming composites. Thus, these results highlight the peculiar nature of parallel indicators. To avoid being deceived by this special case, researchers studying factor analysis and related methods need to vary parameter values for the different indicators of a given factor.

Empirical example 2: simulated data with unequal loadings

Method

This example uses the same model structure but with loadings that vary within each common factor. The three factors use different sets of standardized loadings for their three indicators (0.9/0.7/0.5 for the first common factor, 0.8/0.6/0.4 for the second, and 0.9/0.6/0.3 for the third) to avoid the parallel indicator situation. Variances for the specific factors were again set so that each observed variable had unit variance. Thus, for each common factor, some of its indicators have stronger or weaker correlations with other indicators across the model—but the proportionality constraints underlying the factor model still hold (Jöreskog, 1979).

Results

Again, given the specified common factor model and the absence of random sampling variance, the lavaan package returned parameter estimates that were exactly equal to the programmed values. In this situation (see Table 3), with *standardized* composites, RCA parameter estimates are equal to those for PLS path modeling and GSCA, but only for PLS_B and $GSCA_W$. Parameter estimates from PLS_A and $GSCA_{WL}$ are *not* equal to RCA estimates, not equal to PLS_B or $GSCA_W$ estimates, and not exactly equal to each other, even in this situation where random sampling variance has been completely eliminated. This seems to confirm that $GSCA_W$ weights are, in fact, regression weights, while $GSCA_{WL}$ weights, like PLS_A weights, are correlation weights. In this empirical example, the use of correlation weights substantially conflicts with the common factor model. Estimation under either PLS_A or $GSCA_{WL}$ struggles to find an optimal solution in the face of contrary evidence—variation in correlations across observed variable predictors. Estimation under PLS_B and $GSCA_W$ uses all available information and matches the weighted composite solution offered by RCA and, of course, by regression method factor scores.

Table 3. Parameter estimates from empirical example 2: Unequal loadings.

	Regression component analysis			GSCA and PLS path modeling			
	FA	RCA_F	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
Loadings							
y1	0.900		0.978	0.978	0.978	0.905	0.887
y2	0.700		0.760	0.760	0.760	0.833	0.829
y3	0.500		0.543	0.543	0.543	0.665	0.698
y4	0.800		0.945	0.945	0.945	0.873	0.853
y5	0.600		0.709	0.709	0.709	0.775	0.776
y6	0.400		0.473	0.473	0.473	0.574	0.610
y7	0.900		0.987	0.987	0.987	0.911	0.873
y8	0.600		0.658	0.658	0.658	0.796	0.806
y9	0.300		0.329	0.329	0.329	0.454	0.524
Weights							
y1		0.852	0.785	0.785	0.785	0.520	0.475
y2		0.247	0.227	0.227	0.227	0.405	0.412
y3		0.120	0.110	0.110	0.110	0.289	0.340
y4		0.878	0.743	0.743	0.743	0.574	0.536
y5		0.370	0.314	0.314	0.314	0.431	0.438
y6		0.188	0.159	0.159	0.159	0.287	0.333
y7		0.962	0.877	0.877	0.877	0.628	0.547
y8		0.190	0.174	0.174	0.174	0.419	0.458
y9		0.067	0.061	0.061	0.061	0.209	0.294
Structural Paths							
F1->F2		0.390		0.390	0.390	0.373	0.365
F2=>F3		0.386		0.386	0.386	0.366	0.353

FA: factor analysis; RCA_F: regression component analysis with factor standardization; RCA_C: regression component analysis with composite standardization; PLS_B: partial least squares path modeling mode B; $GSCA_W$: generalized structured component analysis with weights only; PLS_A: partial least squares path modeling mode A; $GSCA_{WL}$: generalized structured component analysis with weights and loadings.

Empirical example 3: composite model population

Method

Cho and Choi (2020) describe a procedure for creating data that specifically conforms to the model estimated by GSCA. For a given group of observed variables, the composite of interest is defined by the first principal component for the group's correlation matrix, rather than by a common factor. In Cho and Choi's (2020) procedure, residual covariances among the observed variables associated with each different composite are specified such that loadings (again, zero-order correlations between observed variable and composite) are consistent with correlations among observed variables associated with each separate composite. Only this first principal component, for each group of observed variables, can correlate with other composites. Then the correlation matrix is assembled essentially as specified in Equation (2), except that "$\mathbf{\Phi}$" in this approach is a matrix of specified correlations among those first principal components, making them canonical variables. This procedure ensures that the GSCA optimization method will reproduce the specified weights, loadings, and composite correlations. This procedure produces a population correlation matrix that does *not* conform to a common factor model. These correlated first principal components carry correlation across groups of observed variables (see Table 4 for the step-by-step process) in much the same way that common factors carry correlation in a common factor model.

This example used the same structural parameter values as in the two previous examples—correlations among the composites match the correlations among common factors in the previous examples. Correlations among the observed variables associated with each principal component match the correlations from the second empirical example.

This demonstration used the mvrnorm() function within the MASS package in R to simulate 50,000 cases consistent with the observed variable correlation matrix developed for this example. To demonstrate that the results reported here are not an artifact of the Bollen-Stine adjustment, this example used the "empirical = TRUE" option associated with the mvrnorm() function to obtain simulated data with moments that exactly match those specified (see Table 5). A common factor model was estimated using the lavaan package, then weights were computed using Equation (7), using the factor-model implied observed variable correlation matrix for Σ. Then composite scores were estimated, their covariances and correlations computed, and then standardized weights were computed.

Table 4. Constructing data for a composite-based model, step by step.

Step 1: Specify the covariance matrix among the observed variables associated with each of the *K* composites.
Step 2: Specify the weights linking each composite to its associated observed variables, then standardize to ensure that the composite will have unit variance, given the specified covariances among the observed variables.
Step 3: Given the specified weighted composites, calculate the loading of each observed variable on its composite as a zero-order correlation, and assemble the loading matrix $\mathbf{\Lambda}$.
Step 4: Calculate residual covariances among each composite's observed variables and assemble the model-wide residual covariance matrix $\mathbf{\Psi}$.
Step 5: Specify the structural model between composites and calculate the covariance matrix of the composites $\mathbf{\Phi}$.
Step 6: Calculate the observed variable covariance matrix $\mathbf{\Sigma} = \mathbf{\Lambda}\mathbf{\Theta}\mathbf{\Lambda}' + \mathbf{\Psi}$.

Table 5. Observed variable correlation matrix for empirical example 3.

	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9
y_1	1.000								
y_2	0.630	1.000							
y_3	0.450	0.350	1.000						
y_4	0.361	0.343	0.293	1.000					
y_5	0.342	0.325	0.278	0.480	1.000				
y_6	0.281	0.267	0.228	0.320	0.240	1.000			
y_7	0.187	0.177	0.152	0.352	0.333	0.274	1.000		
y_8	0.178	0.169	0.144	0.335	0.318	0.261	0.540	1.000	
y_9	0.120	0.114	0.097	0.226	0.214	0.176	0.270	0.180	1.000

Results

From the lavaan results, the χ^2 statistic's value ($\chi^2 = 3660$, $df = 25$, $p < .001$) affirms that the data do not conform to the estimated common factor model. In empirical examples 1 and 2, the χ^2 statistic's value was 0 due to the absence of both specification error and random sampling variance. RCA assumes that a factor model is correct in the population, but here it is not.

There are two major differences between these results and those from the prior empirical examples. First, the RCA weights and loadings do not match those of any of the forms of PLS path modeling or GSCA, even after the RCA composites are standardized (see Table 6). Discrepancies in standardized parameter estimates are as large as 0.200. The second major difference is that PLS path modeling and GSCA results are all identical to each other, regardless of form—there is no distinction between regression weight forms and correlation weight forms. In addition, PLS path modeling and GSCA exactly reproduce the values (0.500) for structural parameters used to build the correlation matrix, while estimates from RCA only approximately match those values.

A Bollen-Stine type transformation (as in Equation (22)) forced the dataset to fit the estimated common factor model. The model-implied correlation matrix

Table 6. Parameter estimates from empirical example 3: composite-based population.

	Regression component analysis			GSCA and PLS path modeling			
	FA	RCA_F	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
Loadings							
y1	0.842		0.941	0.876	0.876	0.876	0.876
y2	0.739		0.826	0.831	0.831	0.831	0.831
y3	0.527		0.589	0.711	0.711	0.711	0.711
y4	0.724		0.872	0.825	0.825	0.825	0.825
y5	0.656		0.790	0.782	0.782	0.782	0.782
y6	0.461		0.555	0.642	0.642	0.642	0.642
y7	0.768		0.910	0.853	0.853	0.853	0.853
y8	0.691		0.818	0.813	0.813	0.813	0.813
y9	0.341		0.404	0.548	0.548	0.548	0.548
Weights*							
Y1		0.719	0.644	0.446	0.446	0.446	0.446
y2		0.404	0.362	0.423	0.423	0.423	0.423
y3		0.181	0.162	0.362	0.362	0.362	0.362
y4		0.715	0.594	0.484	0.484	0.484	0.484
y5		0.542	0.450	0.459	0.459	0.459	0.459
y6		0.276	0.229	0.377	0.377	0.377	0.377
y7		0.754	0.637	0.505	0.505	0.505	0.505
y8		0.532	0.449	0.482	0.482	0.482	0.482
y9		0.156	0.131	0.324	0.324	0.324	0.324
Structural Paths							
F1->F2			0.484	0.500	0.500	0.500	0.500
F2=>F3			0.484	0.500	0.500	0.500	0.500

FA: factor analysis; RCA_F: regression component analysis with factor standardization; RCA_C: regression component analysis with composite standardization; PLS_B: partial least squares path modeling mode B; $GSCA_W$: generalized structured component analysis with weights only; PLS_A: partial least squares path modeling mode A; $GSCA_{WL}$: generalized structured component analysis with weights and loadings.

Table 7. Parameter estimates from empirical example 3: transformed data.

	Regression component analysis			GSCA and PLS path modeling			
	FA	RCA_F	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
Loadings							
y1	0.842		0.941	0.941	0.941	0.886	0.875
y2	0.739		0.826	0.826	0.826	0.847	0.841
y3	0.527		0.589	0.589	0.589	0.686	0.710
y4	0.724		0.878	0.878	0.878	0.836	0.829
y5	0.656		0.795	0.795	0.795	0.801	0.796
y6	0.461		0.559	0.559	0.559	0.634	0.650
y7	0.768		0.910	0.910	0.910	0.867	0.848
y8	0.691		0.818	0.818	0.818	0.835	0.825
y9	0.341		0.404	0.404	0.404	0.500	0.552
Weights							
y1		0.719	0.644	0.644	0.644	0.486	0.462
y2		0.404	0.362	0.362	0.362	0.426	0.420
y3		0.181	0.162	0.162	0.162	0.304	0.341
y4		0.715	0.590	0.590	0.590	0.509	0.498
y5		0.542	0.447	0.447	0.447	0.461	0.455
y6		0.276	0.227	0.227	0.227	0.324	0.346
y7		0.754	0.637	0.637	0.637	0.544	0.513
y8		0.532	0.449	0.449	0.449	0.489	0.481
y9		0.156	0.131	0.131	0.131	0.242	0.304
Structural paths							
F1->F2			0.459	0.459	0.459	0.451	0.448
F2=>F3			0.459	0.459	0.459	0.453	0.448

FA = factor analysis. RCA_F = regression component analysis with factor standardization. RCA_C = regression component analysis with composite standardization. PLS_B = partial least squares path modeling mode B. $GSCA_W$ = generalized structured component analysis with weights only. PLS_A = partial least squares path modeling mode A. $GSCA_{WL}$ = generalized structured component analysis with weights and loadings.

from the original factor analysis of this data provided the "target" correlation matrix. The result is a dataset of 50,000 observations, with an assortment of loading values, that follows the common factor model with no random sampling variance. Estimating the common factor model using this transformed dataset produced the expected χ^2 value of 0, with 25 degrees of freedom.

In an analysis of this transformed data, RCA results with standardized composites matched those of PLS_B and $GSCA_W$, just as in the second empirical example (see Table 7). Results from PLS_A and $GSCA_{WL}$ differ from these numbers and do not match each other exactly, again even in the absence of random sampling variance. Thus, the pattern of results with the data transformed to fit a factor model matches the pattern observed in Empirical Example 2.

Empirical example 4: real data

Method

This example uses data distributed with the cSEM R package and used in documentation for Hwang et al. (2021) stand-alone GSCAPro package. Henseler (2021, p. 135) attributes the items in the scale to Bergami

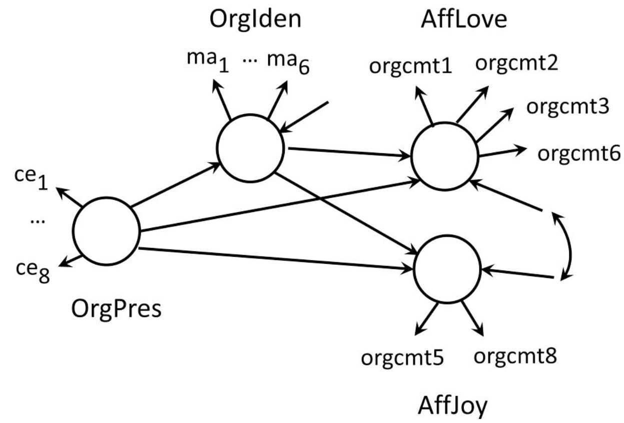

Figure 2. Path model for empirical example 4.

and Bagozzi (2000). The dataset shared *via* cSEM includes 305 cases and 22 variables. A dichotomous gender variable is excluded here because the dichotomy is incompatible with standardization. The other 21 variables are based on five-category response scales, so multivariate normality will not strictly hold. Otherwise, this example specifies the same model presented by Rademaker and Schuberth (2022, p. 11) (see Figure 2), which includes only 20 of the remaining 21 observed variables. The model includes a free residual covariance between the structural error terms for the common factors labeled "AffLove" and "AffJoy." PLS path modeling and GSCA do not include residual structural covariances as model parameters, but (as a reviewer correctly pointed out) this means that such covariances are not constrained within those approaches.

Results

The mardia() function in the R package psych (version 2.3.3) calculated the value of the Mardia (1970) multivariate kurtosis statistic for the 20 observed variables in the model (b_{2p} = 573.7, kurtosis = 39.4, $p <$ .001). This indicated excess multivariate kurtosis, which may have contributed to the poor fit for the maximum likelihood (ML) solution for this model ($\chi^2 = 635.0$, $df = 164$, $p < 0.001$). Robust variants available with the lavaan package yielded somewhat smaller χ^2 values (MLR scaled $\chi^2 = 495$, $p < .001$; MLM scaled $\chi^2 = 423$, $p < .001$) but with the same parameter estimates as ordinary ML.

Diagonally weighted least squares (DWLS) estimation suggests an extremely good fit ($\chi^2 = 112$, $df = 164$, $p = .999$) and somewhat different parameter estimates. This presents an opportunity to evaluate the impact of the distributional assumption on the

Table 8. Parameter estimates from empirical example 4: original data.

Average loading	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
OrgPress	0.646	0.773	0.770	0.800	0.800
OrgIden	0.659	0.749	0.745	0.760	0.760
AffLove	0.640	0.764	0.763	0.767	0.767
AffJoy	0.700	0.835	0.836	0.841	0.842
Weights Root Mean Squared Difference	RCA_C v PLS_B	RCA_C v $GSCA_W$		RCA_C v PLS_A	RCA_C v $GSCA_{WL}$
OrgPres (8)	0.14196	0.14667		0.05165	0.05363
OrgIden (6)	0.08759	0.09892		0.04760	0.04719
AffLove (4)	0.08066	0.07994		0.07295	0.07191
AffJoy (2)	0.11587	0.11632		0.13776	0.15981
Structural Parameter Estimates for Standardized Composites)	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
OrgPres > OrgIden	0.360	0.350	0.346	0.360	0.360
OrgPres > AffLove	0.188	0.197	0.198	0.195	0.194
OrgIden > AffLove	0.542	0.552	0.553	0.543	0.543
OrgPres > AffJoy	−0.128	−0.157	−0.159	−0.124	−0.120
OrgIden > AffJoy	−0.370	−0.371	−0.373	−0.363	−0.364

RCA_F: regression component analysis with factor standardization; RCA_C: regression component analysis with composite standardization; PLS_B: partial least squares path modeling mode B; $GSCA_W$: generalized structured component analysis with weights only; PLS_A: partial least squares path modeling mode A; $GSCA_{WL}$: generalized structured component analysis with weights and loadings.

phenomena examined here. The prior example established that RCA results differ substantially from PLS and GSCA results when the factor model does not fit. Do RCA results based on the DWLS solution match the PLS path modeling and GSCA results?

Unfortunately, no (see Table 8). Because of the large number of parameter estimates, only the average loading for each composite is reported. The average of standardized *weights* is necessarily constrained—the more items included in a standardized composite, the smaller the average individual weight must be, assuming the average covariances between items do not change. Therefore, instead of reporting average weight per composite, this paper reports root mean square differences in weights per item between RCA and each of the other four approaches (see Table 8). Granted, root mean square difference captures both systematic differences between estimates and variability of estimates for the different techniques, but it still provides some basis for comparison. Structural parameter estimates for the composite models are also reported.

In fact, neither RCA solution—RCA based on ML estimates or RCA based on DWLS estimates—produced a close match overall to the parameter estimates provided by any form of PLS path modeling or GSCA, even with standardized composites, and even though the DWLS results implied a good fit. Structural parameter estimates are generally similar, but mean square differences in weights are fairly large, and differences in average loadings appear substantial. In contrast, despite the presence of random sampling variance and differences in specific estimation algorithms, results are very similar across PLS and GSCA, especially for PLS_B and $GSCA_W$ and for PLS_A and $GSCA_{WL}$.

As in Empirical Example 3, the Bollen-Stine transformation (Equation (22)) was applied, using the model-implied covariance matrix from the first factor analysis of these data as the transformation's target. Thus, the data were transformed to exactly match the estimated common factor model. However, the transformation did not change the excess kurtosis of the original data, though the transformed data produce a χ^2 value of 0 for the factor model, again using lavaan's DWLS estimator.

Similar to the results for Empirical Example 3 with transformed data, RCA results *almost* exactly match results from PLS_B and $GSCA_W$ (see Table 9). Discrepancies are quite small, especially in comparison to the results reported in Table 8. Results for PLS_A and for $GSCA_{WL}$ match each other almost exactly, as do results for PLS_B and $GSCA_W$. Root mean square differences in weights are much smaller. For PLS_B and $GSCA_W$, none of the root mean square differences is as large as 0.00001, while differences between RCA weights and those for PLS_A and $GSCA_{WL}$ are much larger. Structural model parameter estimates are quite similar across all methods, though the affinities between RCA, PLS_B and $GSCA_W$, on the one hand, and PLS_A and $GSCA_{WL}$, on the other, are evident.

Table 9. Parameter estimates from empirical example 4: transformed data.

Average Loading	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
OrgPress	0.794	0.794	0.794	0.795	0.795
OrgIden	0.756	0.756	0.756	0.758	0.758
AffLove	0.766	0.766	0.766	0.767	0.767
AffJoy	0.836	0.836	0.836	0.842	0.842
Weights Root Mean Squared Difference	RCA_C v PLS_B	RCA_C v $GSCA_W$		RCA_C v PLS_A	RCA_C v $GSCA_{WL}$
OrgPres (8)	0.00000	0.00000		0.02979	0.03070
OrgIden (6)	0.00000	0.00000		0.03099	0.03100
AffLove (4)	0.00000	0.00000		0.02607	0.02774
AffJoy (2)	0.00000	0.00000		0.06993	0.09785
Structural Parameter Estimates for Standardized Composites)	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
OrgPres > OrgIden	0.357	0.357	0.357	0.356	0.356
OrgPres > AffLove	0.198	0.198	0.198	0.198	0.198
OrgIden > AffLove	0.541	0.541	0.541	0.540	0.540
OrgPres > AffJoy	−0.128	−0.128	−0.128	−0.128	−0.127
OrgIden > AffJoy	−0.361	−0.361	−0.361	−0.360	−0.359

RCA_F: regression component analysis with factor standardization; RCA_C: regression component analysis with composite standardization; PLS_B: partial least squares path modeling mode B; $GSCA_W$: generalized structured component analysis with weights only; PLS_A: partial least squares path modeling mode A; $GSCA_{WL}$: generalized structured component analysis with weights and loadings.

Summary

The empirical examples here demonstrate properties of Schönemann and Steiger's (1976) regression component analysis, a composite-based method that predates the literatures of today's better-known composite-based methods, partial least squares (PLS) path modeling and generalized structured component analysis (GSCA). Trivially, results from regression component analysis based on common factor standardization (RCA_F) are equivalent to factor analysis results using the same standardization, because RCA_F retains factor model parameter estimates, supplemented by a weight matrix computed from factor model parameter estimates.

The first example showed that, when a factor model holds exactly (without either random sampling variance or misspecification) and when the indicators for each common factor are parallel, regression component analysis with composite standardization (RCA_C), PLS and GSCA yield identical solutions. Results are equivalent regardless of whether the PLS or GSCA approach includes only weights (PLS_B, $GSCA_W$) or both weights and loadings (PLS_A, $GSCA_{WL}$). The difference in standardization between the factor-standardized results (factor analysis, RCA_F) and the composite-standardized results (RCA_C, PLS_B, PLS_A, $GSCA_W$, $GSCA_{WL}$) draws a persistent if cosmetic dividing line between the two classes of approaches, across all examples considered here (see Table 10).

The second example showed that, when the factor model holds exactly but each factor's indicators are only congeneric, results from RCA_C are only equivalent to PLS_B and $GSCA_W$. Results from PLS_A and $GSCA_{WL}$ are no longer identical, either to RCA_C results or to each other's, though they are generally similar. The regression weights used by RCA_C, PLS_B and $GSCA_W$ make use of the correlations between the indicators for each composite, while the correlation weights in PLS_A and $GSCA_{WL}$ do not.

The third example demonstrated the dependence of regression component analysis on the common factor model. This example constructed a population based on composites, not common factors, so the factor model was misspecified but the data were still free of random sampling variance. In this example, RCA_C results, rescaled to standardize the composites but still anchored to the factor model results through the original loadings and weights, were different from results from all versions of PLS and GSCA, which were all identical to each other. However, when a Bollen-Stine type transformation was imposed on the data, so that a common factor model did hold, then the outcome matched that from the second example, with results from RCA_C, PLS_B and $GSCA_W$ all identical, while results from PLS_A and $GSCA_{WL}$ were different, and different from each other, though generally similar.

The fourth example used a real data set with a finite sample size—hence, with substantial random sampling variance—and high excess kurtosis. The common factor model fit poorly under maximum likelihood (ML) estimation and robust variants but yielded a low χ^2 using the lavaan package's "diagonally weighted least squares" (DWLS) estimator.

Table 10. Summary of results.

1. Zero Discrepancy Factor Model, Parallel Indicators	FA	RCA_F	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
2. Zero Discrepancy Factor Model, Congeneric Indicators	FA	RCA_F	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
3a. Zero Discrepancy Composite Model	FA	RCA_F	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
3b. Transformed to Zero Discrepancy Factor Model, Congeneric Indicators	FA	RCA_F	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
4a. Real Data, Factor Model Poor Fit	FA	RCA_F	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$
4b. Transformed to Zero Discrepancy Factor Model, Congeneric Indicators	FA	RCA_F	RCA_C	PLS_B	$GSCA_W$	PLS_A	$GSCA_{WL}$

FA: factor analysis; RCA_F: regression component analysis with factor standardization; RCA_C: regression component analysis with composite standardization; PLS_B: partial least squares path modeling mode B; $GSCA_W$: generalized structured component analysis with weights only; PLS_A: partial least squares path modeling mode A; $GSCA_{WL}$: generalized structured component analysis with weights and loadings. Solid-line boxes indicate no parameter estimates differences larger than 0.001. Dashed-line boxes indicate no parameter estimates differences larger than 0.005.

Regression component analysis results were similar, though different, across ML and DWLS estimation of the factor model. As in the third example, RCA_C results were different from the results from either form of PLS or GSCA. Results for PLS_B nearly matched results for $GSCA_W$, while results for PLS_A nearly matched results for $GSCA_{WL}$. Transformation of the data, which left the excess kurtosis intact, produced RCA_C results that almost exactly matched results from PLS_B and $GSCA_W$, which were essentially identical to each other. Results from PLS_A and $GSCA_{WL}$ were slightly different but also essentially matched each other.

Conclusions and implications

It is clearly possible to replace the common factors in a structural equation model with weighted composites of the observed variables, consistent with the same data set. One can do this by first estimating the factor-based model and then calculating a weight matrix using factor model parameter estimates and the model-implied observed variable covariance matrix, thus using no additional information. Like regression method factor scores, regression component analysis composites have the same covariances as the original common factors but different variances, so standardized coefficients linking the composites are different from standardized coefficients linking the factors, while unstandardized coefficients are identical. Some readers will be reminded of how ignoring random measurement error produces similar differences between unstandardized and standardized parameter estimates in regression or path analysis (Rigdon, 1994). Recall that "error terms" are still a part of the regression component analysis model. However, while specific factors in factor analysis are orthogonal to the common factors, observed variable "error terms" in regression component analysis are not orthogonal to the composites which the observed variables form.

When the factor model is correct in the population, RCA_C (composite standardization) parameter estimates match estimates obtained *via* PLS_B and $GSCA_W$—which all use regression weights—but not, in general, estimates obtained *via* PLS_A or $GSCA_{WL}$—which employ correlation weights. It seems ironic that this equivalence applies to the regression weight versions of PLS path modeling and GSCA, not to the correlation weight versions which are more similar in apparent form to regression component analysis and which can be transformed or extended to obtain consistent estimates of factor model parameters. The fundamental problem is that if a factor model holds, the observed variables loading on the same common factor will be correlated—ideally, highly correlated—so using correlation weights to build composites ignores much information. In the special case of parallel indicators in the common factor model, this difference is obscured, and standardized composites under both regression weights and correlation weights produce the same solution, matching RCA_C results. Overall, though, the results reported here suggest that RCA, PLS_W and $GSCA_W$ are all consistent approaches for modeling data that conforms to a factor model.

What is a practical researcher to make of these results? Regression component analysis avoids the indeterminacy inherent in common factor solutions, but that indeterminacy does not affect parameter estimates or model fit, and factor model users are inclined to dismiss it. Suppose a factor model holds, and the researcher is only interested in parameter estimates and the model's fit to the moments of a population. In that case, they do not need to consider any method other than factor analysis. It is certainly true that both PLS path modeling and GSCA include extensions to recover factor model parameters. However, there seems no reason to take such a roundabout approach if what the researcher wants in the first place is a factor model solution.

If a factor model holds but the researcher has reason to require specific scores for each respondent, as perhaps in personnel decisions or legal proceedings, RCA (with either standardization) provides an immediate and easily implemented solution. Yes, the resulting scores are not true factor scores because they omit such scores' arbitrary random component. However, that omission can be an advantage in some situations, and this approach is relatively easy to explain, given the original factor model. Researchers can obtain essentially the same results using the regression weight versions of PLS path modeling or GSCA, but that route involves extra complications without obvious advantages.

It could be argued that, in such a situation, a researcher might simply estimate the factor model and then calculate "regression method" factor scores, and bypass RCA altogether. The difference, though, is that "regression method" factor scores are mathematically inconsistent with the factor model from which they are derived, while the same scores are entirely consistent with the RCA model. Thus, adopting the RCA model eliminates this discrepancy.

If a factor model does *not* hold—or holds for some sets of observed variables but not for others—then extensions of PLS path modeling or GSCA come to the fore. In particular, Hwang et al., (2021) "Integrated Generalized Structured Component Analysis" enables the researcher to make different choices for different sets of observed variables, associating some with a common factor and others with a composite. The Henseler-Ogasawara specification allows similar flexibility, and the model can be estimated using flexible factor model software (Schuberth, 2023). RCA and factor analysis itself are not good choices in such a case because both rely on the assumption that the factor model is correct in the population across the whole model.

McDonald's (1996, p. 251) path-breaking analysis asserted that PLS path modeling with loadings "is equivalent in an obvious sense to a factor model with a single inter-battery factor accounting for all correlations of directly connected blocks." Tucker's (1958) inter-battery factor model involved different sets (or batteries) of observed variables loading on the *same* common factor. PLS path modeling associates multiple sets of observed variables with *different* distinct composites. Browne (1979, p. 82) made a similar point while demonstrating the close connection between the inter-battery factor model and canonical correlation analysis. Yes, the free residual covariances among the observed variables within each set, a feature of the inter-battery factor model, is also consistent with the GSCA and PLS path models, in contrast to conventional constrained factor models. Still, it appears more correct to identify the regression weight forms of GSCA and PLS with regression component analysis and with "regression method" factor scores (Yuan & Deng, 2021; but see Schuberth et al., 2023; Yuan & Deng, 2023).

These empirical examples also illustrate the essential equivalence of GSCA and PLS path modeling when both methods use regression weights, and an approximate equivalence when both use correlation weights. Results here support conclusions drawn from Cho and Choi's (2020) simulations. Other recent research (Cho et al., 2022), limited to PLS_A and $GSCA_{WL}$, suggests a small advantage in performance for GSCA. Regarding published applications, PLS path modeling seems far more popular, but it is unclear why that should continue to be true. GSCA has already provided the framework for a host of extensions (Hwang et al., 2017, 2021; Jung et al., 2012). GSCA's compact and unified statistical foundation provides a robust basis for further extensions. GSCA can estimate models where observed variables associate with multiple composites, rather than just one (Cho et al., 2022). As noted here, software exists that can use either approach. It seems obvious that GSCA is not getting the consideration that it deserves.

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